



Saint Columban College  
College of Computing Studies  
Pagadian City



## File-Run1

file-run1 



Medium Reverse Engineering picoCTF 2022

AUTHOR: WILL HONG

### Description

A program has been provided to you, what happens if you try to run it on the command line?

Download the program [here](#).

Hints 

1 2

Try running it by adding a '.' in front of the path to the file (i.e. `$ ./run`)

26,806 users solved



63% Liked



picoCTF{FLAG}

Submit Flag

**Author:** The Analyst: Hyposelenia

**Challenge:** File-Run1

**Category:** Reverse Engineering

**Date:** 2/26/26



## I. Objective

The objective of this challenge is to understand how to **identify and execute binary files in Linux**.

The goal is to recognize that the downloaded file is an executable program and run it to **retrieve the flag**.

## II. Background

In Linux, files are not always opened like in Windows (double-click). Instead, files may require **execution permissions** before they can run.

Executable files (especially in CTFs) are often **compiled programs** that must be run through the terminal.

Key concepts involved:

- File permissions
- Executable files
- Running programs via terminal

## III. Tool Used

- **Kali Linux (Oracle VirtualBox)** - Environment used to run commands

## IV. Methodology

1. Launched **Kali Linux** using Oracle VirtualBox.



2. Downloaded the file using:

*wget <link\_address>*

```
(kali㉿kali)-[~/Downloads]
$ wget https://artifacts.picoctf.net/c/218/run
--2026-03-26 05:18:55-- https://artifacts.picoctf.net/c/218/run
Resolving artifacts.picoctf.net (artifacts.picoctf.net)... 18.172.2
Connecting to artifacts.picoctf.net (artifacts.picoctf.net)|18.172.
HTTP request sent, awaiting response... 200 OK
Length: 16736 (16K) [application/octet-stream]
Saving to: 'run'

run
100%[=====]
2026-03-26 05:18:56 (652 KB/s) - 'run' saved [16736/16736]
```

3. Listed files in the directory:

*ls*

```
(kali㉿kali)-[~/Downloads]
$ ls
cat.jpg leak leak.tar run
```

4. Observed the file (displayed in green, indicating it may be executable or intended to be executable).

*run*

5. Added execute permission using:

*chmod +x <filename>*

```
(kali㉿kali)-[~/Downloads]
$ chmod +x run
```

6. Executed the file using:

*./<filename>*

```
(kali㉿kali)-[~/Downloads]
$ ./run
The flag is: picoCTF{U51N6_Y0Ur_F1r57_F113_9bc52b6b}
```



7. The program ran and displayed the **flag**.

## V. Result

picoCTF{U51N6\_Y0Ur\_F1r57\_F113\_9bc52b6b}

```
picoCTF{U51N6_Y0Ur_F1r57_F113_9bc52b6b}
```

## VI. Explanation

### What is an Executable File?

An executable file is:

A file that can be **run as a program**

### Real-life analogy:

- A normal file = a **book you read**
- An executable file = a **machine you turn on**

### What does `chmod +x` do?

Command:

*`chmod +x <filename>`*

### Breakdown:

- `chmod` = change mode (permissions)
- `+x` = add execute permission

### Simple meaning:

*“Allow this file to be run as a program”*

### What does `./<filename>` mean?

Command:

*`./program`*

### Breakdown:



- . → current directory
- / → access file
- program → file name

Meaning:

*“Run this file located in the current folder”*

### Simple Summary

- File = executable program
- chmod +x = allow execution
- ./file = run program
- Program outputs flag

## VII. Conclusion

This challenge introduces the basics of working with **executable files in Linux**.

By understanding file permissions and how to execute programs from the terminal, the flag was successfully retrieved.

This reinforces an essential skill in cybersecurity:

**Knowing how to properly run and interact with files is the first step in reverse engineering and binary analysis.**

— The Analyst: Hyposelenia